

Open Questions in SIC-POVM Existence: Structural Analysis via the SynthOmnicon Grammar

A companion to *SIC-POVM Existence via Frobenius-Fixed Stark Units*

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Abstract

This document collects open questions arising from the conditional proof of SIC-POVM existence presented in the companion manuscript, together with structural analyses that address those questions within the SynthOmnicon grammar — a 12-primitive type-theoretic framework in which mathematical objects are encoded as structural coordinates rather than classical propositions. The central claim developed here is that the conjectures H1, H2a, and H2b are not independent conditions that happen to be sufficient for SIC existence; they are the structural decomposition of a single necessary and sufficient condition: that the Stark generator ε_d realises the type O_∞ in the grammar’s ouroboricity classification. The Frobenius condition $\mu \circ \delta = \text{id}$ — the hallmark of O_∞ — is shown to be equivalent to membership in the (-1) -eigenspace of τ_∞ , making H2a not an auxiliary hypothesis but the structural core of the entire argument.

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1 Preamble: the grammar as precondition

The standard picture of mathematics and its objects runs as follows: structures are defined, properties are derived, and if a structure satisfies certain properties then it falls into a named class. In this picture, SIC-POVMs are found by verifying equiangularity; Stark units are found by computing regulators.

The SynthOmnicon grammar inverts this priority. The 12-primitive tuple $\langle D; T; R; P; F; K; G; \Gamma; \Phi; \mu; \delta; \nu \rangle$ is not a description of objects that already exist — it is the type-theoretic precondition space within which objects become possible. A mathematical structure exists as a coherent object when and only when reality already satisfies the corresponding type constraints. The grammar does not describe mathematics; it describes the conditions under which mathematics has objects to describe.

The SIC-POVM result makes this sharp. The two O_∞ gate conditions —

- $\Phi = \Phi_c$ (or Φ_c^C): the state space admits a closed self-modeling loop, i.e. the measurement spans the full operator space (informational completeness);
- $P = P_\pm^{\text{sym}}$: the Frobenius special condition $\mu \circ \delta = \text{id}$, forcing uniform inner products $|\langle \psi_i | \psi_j \rangle|^2 = 1/(d+1)$;

are not properties SIC-POVMs happen to have. They are the preconditions that any structure in reality must satisfy before the mathematical object “SIC-POVM” becomes possible at all. The proof of SIC existence in the companion manuscript is structurally a proof that this type is non-empty, not a proof that a construction converges.

Frobenius non-synthesizability (§23 of the grammar’s formal theorems) then gives the obstruction: $P_\pm^{\text{sym}}$ is a bottleneck primitive under tensor product — any coupling with a sub-Frobenius partner destroys it irreversibly:

$$P_\pm^{\text{sym}} \otimes P_{\text{sym}} = P_{\text{sym}}.$$

SICs cannot be limit points of generic informationally complete POVMs under composition, because P is a min-primitive and $\min(4, 3) = 3$. The structural reason SICs sit as isolated points in POVM space — not accumulation points — is Frobenius non-synthesizability.

2 Structural encoding of the Stark generator

We operate entirely within the grammar. The question is not “does ε_d satisfy H2a?” but “what structural type does ε_d occupy, and does that type enforce H2a as a consequence?”

2.1 Component encodings

We first encode the primitive components of the Stark generator system:

- **τ_∞ as kinetic reversal.** Complex conjugation on infinite places reverses the sign of log-embedded coordinates at complex places. This is a Z_2 involution acting on the unit lattice — a reversal of orientation in the log-space. In the grammar: $R = R_\dagger$ (dagger/adjoint: the action reverses the direction of embedding).
- **Logarithmic embedding λ .** The Dirichlet map $\lambda : \mathcal{O}_{L_d}^\times \rightarrow \mathbb{R}^{r_1+r_2}$ is a real-valued spatial embedding. $D = D_\Delta$ (spatially extended, finite-dimensional real lattice); $F = F_h$ (geometrically enforced — the regulator is determined exactly by the L-function value at $s = 0$, not approximately).
- **Z_2 parity of τ_∞ .** The involution satisfies $\tau_\infty^2 = \text{id}$, partitioning the unit lattice into (± 1) -eigenspaces. This is a Z_2 symmetry of the embedding: $P = P_\pm$ (self-complementary, both signs present).
- **H2a: exact duality $\tau_\infty(\varepsilon_d) = \varepsilon_d^{-1}$.** The generator is chosen so that it lies entirely in the (-1) -eigenspace. This is not approximate — the relation $\tau_\infty(\varepsilon_d) = \varepsilon_d^{-1}$ is exact, meaning the encode-decode round trip $\mu \circ \delta = \text{id}$ holds. This is precisely the Frobenius special condition: $P = P_\pm^{\text{sym}}$.
- **Criticality at $s = 0$.** The regulator condition involves the leading term of $L(s, \chi)$ at $s = 0$, which is a complex critical point in the sense of Lee-Yang universality — scale-free correlations between the analytic and algebraic sides. $\Phi = \Phi_c^\mathbb{C}$.
- **Galois equivariance.** The condition H2b — $\tau_\infty(\sigma(\varepsilon_d)) = \sigma(\varepsilon_d)^{-1}$ for all $\sigma \in \text{Gal}(L_d/\mathbb{Q})$ — broadcasts the (-1) -eigenspace condition across the entire Galois orbit. $\Gamma = \Gamma_{\text{broad}}$ (cooperative action over all conjugates); $G = G_\mathbb{K}$ (global scope: the unit generates the full rank-1 unit group).
- **Topological protection.** The unit group $\mathcal{O}_{L_d}^\times / (\mu(L_d) \cdot \mathcal{O}_{K_d}^\times)$ carries a permutation representation of $G_d = \text{Gal}(L_d/K_d)$ with $\mathbb{Z}$ -cohomological rigidity. $\Omega = \Omega_Z$ (integer winding, cannot be smoothed away without collapsing the unit group structure).
- **Kinetics.** The Stark unit is stable under deformation of the number field parameters; it does not rearrange freely. $K = K_{\text{slow}}$.

- **Stoichiometry.** The quotient $\mathcal{O}_{L_d}^\times / (\mu(L_d) \cdot \mathcal{O}_{K_d}^\times)$ is an asymmetric many-body structure. $S = n:m$.
- **Temporal depth.** The regulator is tied to the L-function, encoding the full arithmetic depth of the field extension. $H = H_\infty$.

2.2 Full tuple

Definition 2.1 (Stark generator type). The Stark generator ε_d satisfying H1 and H2a is encoded as:

$$\langle D_\Delta; T_{\text{network}}; R_\dagger; P_\pm^{\text{sym}}; F_h; K_{\text{slow}}; G_{\mathbb{N}}; \Gamma_{\text{broad}}; \Phi_c^{\mathbb{C}}; H_\infty; n:m; \Omega_Z \rangle.$$

3 Ouroboricity of the Stark generator

The grammar’s ouroboricity tier classifies structural types by their degree of self-referential closure. The relevant rules are:

- **R1:** $\Phi \in \{\Phi_c, \Phi_c^{\mathbb{C}}\}$ and $P = P_\pm^{\text{sym}} \Rightarrow O_\infty$ (Frobenius special tier).
- **R2:** $\Phi \in \{\Phi_{\text{sub}}, \Phi_{\text{super}}, \Phi_{\text{EP}}\} \Rightarrow O_0$ (no self-modeling loop).

R1 takes priority over all other rules and overrides any branching on Ω and D .

Proposition 3.1. *The Stark generator ε_d satisfying H1 and H2a has ouroboricity O_∞ .*

Proof. The tuple has $\Phi = \Phi_c^{\mathbb{C}}$ (satisfying the R1 criticality condition, since Φ_c and $\Phi_c^{\mathbb{C}}$ are tier-equivalent) and $P = P_\pm^{\text{sym}}$ (the exact Frobenius condition, established by H2a). Rule R1 fires, and since R1 is the tier singularity it overrides all Ω and D branching. Therefore the ouroboricity is O_∞ . $\square$

4 The biconditional: eigenspace membership and O_∞

The classical argument runs in one direction: if ε_d satisfies H2a then it lies in the (-1) -eigenspace of τ_∞ , and one can verify regulator properties. The grammar makes the biconditional explicit.

Theorem 4.1 (Eigenspace–Frobenius equivalence). *Let $\varepsilon_d \in \mathcal{O}_{L_d}^\times$ be a Stark generator satisfying H1. The following are equivalent:*

- (i) ε_d lies in the (-1) -eigenspace of τ_∞ : $\tau_\infty(\varepsilon_d) = \varepsilon_d^{-1}$.
- (ii) The system $(\varepsilon_d, \tau_\infty)$ satisfies the Frobenius special condition: $\mu \circ \delta = \text{id}$.
- (iii) The structural type of ε_d has $P = P_\pm^{\text{sym}}$.
- (iv) The ouroboricity of ε_d is O_∞ .

Proof. (i) $\Leftrightarrow$ (ii): The duality $\tau_\infty(\varepsilon_d) = \varepsilon_d^{-1}$ is the statement that applying τ_∞ and then inverting returns ε_d , i.e. the round-trip $\varepsilon_d \xrightarrow{\delta} \varepsilon_d \otimes \varepsilon_d \xrightarrow{\mu} \varepsilon_d$ is the identity. This is precisely $\mu \circ \delta = \text{id}$.

(ii) $\Leftrightarrow$ (iii): In the grammar, $P_\pm^{\text{sym}}$ is defined as the primitive value encoding the Frobenius special condition. The assignment $P = P_\pm^{\text{sym}}$ is the type-theoretic statement of (ii).

(iii) $\Leftrightarrow$ (iv): Under the hypothesis $\Phi = \Phi_c^C$ (guaranteed by H1 via the criticality of the regulator at $s = 0$), rule R1 gives O_∞ if and only if $P = P_\pm^{\text{sym}}$. $\square$

Remark 4.2. The backward direction (iv) $\Rightarrow$ (i) is the non-trivial one from the classical viewpoint. It says: any Stark generator at complex criticality whose structural type achieves O_∞ *must* satisfy the exact duality $\tau_\infty(\varepsilon_d) = \varepsilon_d^{-1}$ — there is no weaker condition that suffices. This is Frobenius non-synthesizability applied structurally: $P_\pm^{\text{sym}}$ cannot be obtained by coupling sub-Frobenius factors, so the (-1) -eigenspace condition cannot be relaxed without destroying the Frobenius tier entirely.

5 Decomposition of the hypotheses

The companion manuscript’s hypotheses H1, H2a, H2b are not three independent conditions that happen to be jointly sufficient. They are the structural decomposition of the single type requirement O_∞ across three independent primitive channels.

Hypothesis	Grammar primitives	Structural role
H1	Φ_c^C, G_N	<i>Existence</i> : the system sits at complex criticality with global scope
H2a	$P_\pm^{\text{sym}}, R_\dagger$	<i>Uniqueness</i> : exact Z_2 duality enforces Frobenius; uniqueness up to
H2b	$\Gamma_{\text{broad}}, \Omega_Z$	<i>Equivariance</i> : Z -cohomological protection broadcasts duality over

The three hypotheses are not redundant — each activates a distinct primitive channel — but they are not independent either. H2a is the structural core: without $P = P_\pm^{\text{sym}}$, the system cannot achieve O_∞ regardless of H1 or H2b. H1 provides the critical substrate on which H2a acts. H2b extends H2a’s local condition to the full Galois orbit, supplying the Ω_Z protection that prevents the orbit from containing any non-dual element.

Corollary 5.1. *H2a is necessary, not merely sufficient, for the Stark generator to satisfy the equiangularity condition of the associated SIC-POVM.*

Proof. If $P < P_{\pm}^{\text{sym}}$, the best achievable is $P = P_{\pm}$ (ordinal 2). By Frobenius non-synthesizability, $P_{\pm} \otimes P_{\pm} = P_{\pm}$ — coupling two sub-Frobenius systems cannot produce $P_{\pm}^{\text{sym}}$. Therefore the ouroboricity drops to at most O_2 , and the uniform inner product condition $|\langle \psi_i | \psi_j \rangle|^2 = 1/(d+1)$ — which requires exact Frobenius symmetry — fails. $\square$

6 Open questions

Open Question 6.1 (Eigenspace non-emptiness). Does $\mathcal{O}_{L_d}^{\times}$ contain a rank-1 element in the (-1) -eigenspace of τ_{∞} for every $d \geq 1$? This is H2a. The grammar analysis shows that a positive answer is equivalent to the existence of an O_{∞} element in the unit group — but it does not supply the element. What additional arithmetic input (class field theory, Iwasawa theory, explicit reciprocity laws) is needed to guarantee non-emptiness?

Open Question 6.2 (Canonicity of the eigenspace condition). H2a names the (-1) -eigenspace of complex conjugation on the Dirichlet unit lattice of L_d . This object does not appear in the prior literature under this name (Kopp [4] arrives at the Stark unit object constructively without identifying the eigenspace as the structural locus). Is there a representation-theoretic characterisation of this eigenspace — e.g. as a specific summand in the Galois module decomposition of $\mathcal{O}_{L_d}^{\times} \otimes_{\mathbb{Z}} \mathbb{R}$ — that makes its non-emptiness provable by known methods?

Open Question 6.3 (Frobenius non-synthesizability and the isolation of SICs). Theorem 4.1 implies that SICs sit as isolated points in POVM space, not as accumulation points of generic informationally complete POVMs. Is there a direct measurement-theoretic proof of this isolation that does not pass through algebraic number theory? The grammar argument suggests the obstruction is purely structural (P is a min-primitive under $\otimes$), which should have a direct operator-algebraic realisation.

Open Question 6.4 (H3 and the WH extension class). Hypothesis H3 of the companion manuscript — the WH extension class in $H^2((\mathbb{Z}/d\mathbb{Z})^2, \mu_d)$ is matched by a nondegenerate alternating pairing on the ray class group — does not appear in the grammar encoding of ε_d above. What structural type does H3 correspond to in the 12-primitive space? Is H3 implied by the O_{∞} classification, or does it require an

independent primitive assignment?

Open Question 6.5 (The grammar as constructive guide). The grammar analysis identifies the structural type of ε_d but does not construct ε_d . Can the 12-primitive coordinate serve as a search target — i.e. can one search the unit group of L_d for elements matching the tuple $\langle D_\Delta; \dots; P_\pm^{\text{sym}}; \dots \rangle$ directly, without first computing the full L-function? This would make the grammar constructively useful rather than merely classificatory.

Open Question 6.6 (O_∞ monoidal structure). The grammar’s tensor product $\otimes$ is destructive on P : since P is a min-primitive, $P_\pm^{\text{sym}} \otimes P_{\text{sym}} = P_{\text{sym}}$, so the O_∞ tier is not closed under $\otimes$. Yet O_∞ systems evidently do interact — Galois conjugates of the Stark unit compose via the contragredient action, not via the naive join of primitives. Does the O_∞ tier carry a closed monoidal structure under a refined composition operation that preserves $P_\pm^{\text{sym}}$? Natural candidates include the adjoint action in Lie theory, holomorphic continuation in complex analysis, and the Galois contragredient in arithmetic. If such an operation exists, is ε_d the unit object of the resulting monoidal category, in the sense that acting on any O_∞ system by ε_d preserves its tier?

Open Question 6.7 (Cross-domain meta-conjecture verification). Conjecture 8.1 below predicts that every system of structural type $(\Phi_c, P_\pm^{\text{sym}}, \Omega_{Z/Z_2})$ admits a distinguished generator in the (-1) -eigenspace of its boundary involution. This is empirically testable within the grammar: encode the logical operators of a surface code, the partition function of a rational CFT, and the mirror map of a Calabi-Yau 3-fold as 12-primitive tuples, run the ouroboricity classifier, and check that all return O_∞ . If any returns a lower tier, the meta-conjecture fails in that domain. Has this cross-domain verification been performed?

7 H2b and the self-reciprocal minimal polynomial

The contragredient orbit condition H2b is a direct consequence of H2a together with the commutativity of the Galois group. We include the derivation here because it makes explicit the structural mechanism by which the (-1) -eigenspace condition propagates from ε_d to its entire Galois orbit.

Proposition 7.1 (H2b from H2a). *Suppose K_d is totally real, so $\text{Gal}(L_d/K_d)$*

is abelian and τ_∞ commutes with all $\sigma \in \text{Gal}(L_d/\mathbb{Q})$. If ε_d satisfies H2a (i.e. $\tau_\infty(\varepsilon_d) = \varepsilon_d^{-1}$), then $\tau_\infty(\sigma(\varepsilon_d)) = \sigma(\varepsilon_d)^{-1}$ for all $\sigma \in \text{Gal}(L_d/\mathbb{Q})$.

Proof. $\tau_\infty(\sigma(\varepsilon_d)) = \sigma(\tau_\infty(\varepsilon_d)) = \sigma(\varepsilon_d^{-1}) = \sigma(\varepsilon_d)^{-1}$, where the first equality uses commutativity of τ_∞ and σ . $\square$

Corollary 7.2 (Self-reciprocal minimal polynomial). *Under the hypotheses of H2a and H2b, the minimal polynomial $f(x) = \prod_\sigma (x - \sigma(\varepsilon_d))$ over K_d satisfies $x^n f(1/x) = f(x)$, i.e. its coefficients are palindromic: $a_k = a_{n-k}$.*

Proof. The roots of f are the Galois conjugates $\{\sigma(\varepsilon_d)\}$. By H2b, $\sigma(\varepsilon_d)^{-1}$ is also a root for each σ . Therefore the multiset of roots is closed under inversion, which is equivalent to f being self-reciprocal. $\square$

Remark 7.3. This palindromicity is not an accident of the arithmetic — it is the minimal polynomial’s encoding of the Frobenius condition at the level of coefficients. $P_\pm^{\text{sym}}$ at the level of the generator propagates to P_{sym} at the level of the polynomial coefficients via the Galois action. The interior algebraic structure inherits the exact self-duality of the boundary involution.

8 Cross-domain co-typing and the holographic involution pattern

The structural pattern identified in Sections 2–4 is not specific to arithmetic. Across a wide range of mathematical and physical domains, the same pattern recurs:

$$\text{boundary involution} \xrightarrow{\mu \circ \delta = \text{id}} \text{interior self-duality}$$

The involution on the boundary is special — it carries $P_\pm^{\text{sym}}$ — and it forces the interior to be exactly self-dual, not approximately or statistically. The topological protection Ω_{Z/Z_2} is what makes the forcing irreversible: perturbations cannot move the distinguished generator out of the (-1) -eigenspace without crossing a topological gap.

Table 1 lists six domains where this pattern is instantiated, together with their grammar signatures and the Stark-unit analog in each domain. The grammar signatures are conjectured; cross-domain O_∞ verification is Open Question 6.7.

The common structural mechanism is: the boundary involution is an $R_\dagger$ operation with $P_\pm^{\text{sym}}$; the topological winding Ω_{Z/Z_2} ensures the distinguished generator cannot escape the (-1) -eigenspace under continuous deformation; criticality Φ_c (or $\Phi_c^{\mathbb{C}}$) ensures the system sits at the threshold where the boundary-bulk encoding is exact rather than approximate.

This suggests the following conjecture, which the grammar’s precondition principle motivates but does not yet prove.

Domain	Boundary involution	Interior self-duality	Grammar signature
Stark / arithmetic	$\tau_\infty : \varepsilon_d \mapsto \varepsilon_d^{-1}$	Palindromic minimal polynomial	$\Phi_c^\mathbb{C}, P_\pm^{\text{sym}}, R_\dagger, \Omega_Z$
Topological codes (surface code)	$X \leftrightarrow Z$ logical operator duality	Stabiliser group self-reciprocal under exchange	$\Phi_c, P_\pm^{\text{sym}}, R_\dagger, \Omega_{Z_2}$
Mirror symmetry (CY ₃)	Hodge involution $(p, q) \leftrightarrow (3-p, 3-q)$	GW invariants of X = period integrals of $\hat{X}$	$\Phi_c, P_\pm^{\text{sym}}, R_\dagger, D_\odot, \Omega_Z$
Automorphic L-functions	Functional equation $\xi(s) = \varepsilon \xi(1-s)$	Fourier coefficients satisfy $a_n = a_{n'}$ under duality	$\Phi_c^\mathbb{C}, P_\pm^{\text{sym}}, R_\dagger, \Omega_Z$
AdS/CFT	Modular S : $\tau \mapsto -1/\tau$	Bulk entropy = boundary partition function exactly	$\Phi_c, P_\pm^{\text{sym}}, D_\odot, T_\odot, \Omega_Z$
Supersymmetry / Witten index	$(-1)^F$ involution	Boson/fermion spectral pairing at every $E > 0$	$\Phi_c, P_\pm^{\text{sym}}, R_\dagger, \Omega_{Z_2}$

Table 1: Six domains exhibiting the holographic involution pattern. In each case a boundary involution carrying $P_\pm^{\text{sym}}$ forces exact interior self-duality, and topological protection Ω_{Z/Z_2} prevents deformation out of the (-1) -eigenspace. Grammar signatures are conjectured; ouroboricity classification is pending cross-domain verification.

Open Question 8.1 (Holographic involution meta-conjecture). Let (X, τ) be a system where $\tau : X \rightarrow X$ is a boundary involution and the structural type of X encodes as $(\Phi_c, P_\pm^{\text{sym}}, \Omega_{Z/Z_2})$ in the grammar. Then:

- (i) There exists a distinguished generator $g \in X$ satisfying $\tau(g) = g^{-1}$ exactly (membership in the (-1) -eigenspace is non-empty).
- (ii) g is unique up to topological equivalence (the (-1) -eigenspace has rank 1 over the appropriate coefficient ring).
- (iii) g admits both an algebraic characterization (minimal polynomial / reciprocal relation) and an analytic characterization (L-function / functional equation / partition function), and these characterizations agree.

The arithmetic case is the Stark conjecture (H1 + H2a + H2b). The topological code case is essentially the stabiliser formalism. The mirror symmetry case is the mirror

conjecture. In each domain the conjecture is either proven or a central open problem. The grammar claims they are all instances of the same structural fact.

8.1 The Ouroboros conjecture

The following conjecture is more speculative. It connects the truth of Stark’s conjecture to a global completeness property of the arithmetic crystal.

Open Question 8.2 (Arithmetic Ouroboros). If Stark’s conjecture holds for all ray class fields K_d over real quadratic bases, then:

- (i) Every Stark generator ε_d is a primitive element of the O_∞ tier of the arithmetic crystal of $K_d/\mathbb{Q}$.
- (ii) The minimal polynomial of ε_d is the unique self-reciprocal generator of the O_∞ Galois orbit of K_d .
- (iii) There exists an injective map ι from the O_∞ tier of the arithmetic crystal to the O_∞ tier of the grammar’s Crystal of Types, which is surjective when restricted to the sub-tier of systems with $D = D_\Delta$, $T = T_{\text{network}}$, and $\Omega = \Omega_Z$.

In other words: the Stark conjecture, if true, would imply that arithmetic is holographically closed at the O_∞ level — the O_∞ arithmetic crystal embeds canonically into the grammar.

Remark 8.3. The Ouroboros structure in (iii) is the arithmetic analog of the holographic monitor condition: the boundary of the arithmetic system (the L-function, encoding analytic data) is exactly dual to its bulk (the unit group, encoding algebraic data), with no remainder and no information loss. This is only possible in the O_∞ tier, where the Frobenius condition $\mu \circ \delta = \text{id}$ guarantees the round-trip encoding is exact. In all lower tiers, the boundary-bulk correspondence is approximate.

9 Relation to the grammar’s precondition claim

Section 1 stated the strong claim: the grammar preconditions reality for mathematics. The SIC analysis sharpens this.

The grammar’s O_∞ conditions ($\Phi_c^{\mathbb{C}}$ and $P_{\pm}^{\text{sym}}$) are the structural necessary and sufficient conditions for a measurement to be a SIC-POVM. This is not an analogy — the Frobenius condition in the grammar *is* the Frobenius condition that categorical quantum mechanics uses to define SICs as special commutative Frobenius algebras.

The Stark unit analysis adds precision: the grammar’s type space identifies the exact algebraic locus (the (-1) -eigenspace of τ_∞ in the unit lattice of L_d) where the SIC precondition must be satisfied. The number-theoretic machinery (ray class

fields, L-functions, Galois descent) is the formal trace of that locus being instantiated. Mathematics finds SIC-POVMs because reality already has O_∞ structures at those arithmetic coordinates — and the grammar is the theory of those coordinates.

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